

For the use of an oncologist or a hospital or a laboratory only

Prescribing information

CAPIIBINE

Capecitabine Film-Coated Tablets USP 150 mg and

Capecitabine Film-Coated Tablets USP 500 mg

CAUTION: CYTOTOXIC AGENT

COMPOSITION

Capiibine 150

Each film-coated tablet contains:

Capecitabine USP 150 mg

Colours: Titanium Dioxide, Red oxide of Iron and Yellow oxide of Iron

Capiibine 500

Each film-coated tablet contains:

Capecitabine USP 500 mg

Colours: Titanium Dioxide, Red oxide of Iron and Yellow oxide of Iron

Product Description:

Capecitabine Film-Coated tablets USP 150 mg:

Light peach colored, biconvex, oblong film coated tablets with “150” debossing on one side and ‘RDY’ on other side.

Capecitabine Film-Coated tablets USP 500 mg:

Peach colored, biconvex, oblong tablets with “500” debossing on one side and ‘RDY’ on other side.

CLINICAL INFORMATION

Therapeutic Indications:

Breast Cancer:

CAPIIBINE in combination with docetaxel is indicated for the treatment of patients with locally advanced or metastatic breast cancer after failure of cytotoxic chemotherapy. Previous therapy should have included an anthracycline. CAPIIBINE is also indicated as monotherapy for the treatment of patients with locally advanced or metastatic breast cancer after failure of a taxane and an anthracycline-containing chemotherapy regimen or for whom further anthracycline therapy is not indicated.

CAPIIBINE is indicated in combination with lapatinib ditosylate for the treatment of patients with advanced or metastatic breast cancer whose tumors over express HER2 and who have received prior therapy including an anthracycline, a taxane and trastuzumab.

Colon, Colorectal cancer:

CAPRIBINE is indicated for the treatment of patients with metastatic colorectal carcinoma.

CAPRIBINE is indicated as adjuvant treatment of patients following surgery of Stage III (Duke's Stage C) colon cancer.

Oesophagogastric Cancer:

CAPRIBINE is indicated as first-line treatment of patients with advanced oesophagogastric cancer in combination with a platinum-based regimen.

Dosage and method of administration

Standard dosage

CAPRIBINE tablets should be swallowed whole with water within 30 minutes after a meal. CAPRIBINE tablets should not be crushed or cut. If patients cannot swallow CAPRIBINE tablets whole and tablets must be crushed or cut, this should be done by a professional trained in the safe handling of cytotoxic drugs.

Monotherapy:

Colon, Colorectal and breast cancer

The recommended monotherapy starting dose of CAPRIBINE is 1250 mg/m² administered twice daily (morning and evening; equivalent to 2500 mg/m² total daily dose) for 2 weeks followed by a 7-day rest period.

Combination therapy Breast Cancer

In combination with docetaxel

In combination with docetaxel, the recommended dose of CAPRIBINE is 1250 mg/m² twice daily for 2 weeks followed by a 7-day rest period, combined with docetaxel at 75 mg/m² as a 1-hour intravenous infusion every 3 weeks.

Pre-medication according to the docetaxel labelling, should be started prior to docetaxel administration for patients receiving the CAPRIBINE plus docetaxel combination.

In combination with lapatinib ditosylate

In combination with lapatinib ditosylate, the recommended dose of CAPRIBINE is 2000 mg/m²/day administered orally in 2 doses 12 hours apart for 14 days (Day 1-14) in a repeating 21-day cycle combined with lapatinib ditosylate 1250 mg (5 tablets) given orally once daily from Day 1-21. (See manufacturer's prescribing information for lapatinib ditosylate for further information).

Colon, colorectal cancer

In combination with oxaliplatin and/or bevacizumab

In combination with oxaliplatin and/or bevacizumab the recommended dose of CAPRIBINE is 1000 mg/m² twice daily for 2 weeks followed by a 7-day rest period. The first dose of CAPRIBINE is given on the evening of day 1 and the last dose is given on the morning of day 15. Given as a 3-weekly schedule, on day 1 every 3 weeks bevacizumab is administered as a 7.5 mg/kg intravenous infusion over 30-90 minutes followed by oxaliplatin administered as a 130 mg/m² intravenous infusion over 2 hours.

Premedication to maintain adequate hydration and anti-emesis according to the oxaliplatin product information should be started prior to oxaliplatin administration for patients receiving the CAPRIBINE plus oxaliplatin combination. Adjuvant treatment in patients with stage III colon cancer is recommended for a total of 6 months.

Gastric Cancer

In combination with platinum-based regimen:

In combination with a platinum-based compound the recommended dose of CAPRIBINE for the treatment

of advanced gastric cancer is 1000 mg/m² administered twice daily for 14 days followed by a 7-day rest period. The first dose of CAPIIBINE should be given on the evening of day 1 and the last dose should be given on the morning of day 15. If epirubicin is added to this regimen the recommended dose of CAPIIBINE is 625 mg/m² twice daily continuously. Epirubicin at a dose of 50 mg/m² should be given as a bolus on day 1 every 3 weeks. The platinum-based compound (cisplatin at a dose of 60 mg/m² (triple regimen) - 80 mg/m² (double regimen) or oxaliplatin at a dose of 130 mg/m²) should be given on day 1 as a 2 hour intravenous infusion every 3 weeks.

Premedication to maintain adequate hydration and anti-emesis according to the cisplatin/oxaliplatin summary of product characteristics should be started prior to cisplatin/oxaliplatin administration for patients receiving the CAPIIBINE plus cisplatin/oxaliplatin combination.

Dose calculation

CAPIIBINE dose is calculated according to body surface area. The following tables show the standard and reduced dose calculations (see section “Dosage adjustments during treatment”) for a starting dose of CAPIIBINE of either 1250 mg/m² or 1000 mg/m².

Table 1: Standard and reduced dose calculations according to body surface area for a starting dose of CAPIIBINE of 1250 mg/m²

	Dose level 1250 mg/m² (twice daily)				
	Full dose 1250 mg/ m ²	Number of tablets per administration (each administration to be given morning and evening)		Reduced dose (75%) 950 mg/m ²	Reduced dose (50%) 625 mg/ m ²
Body surface area (m ²)	Dose per administration (mg)	150 mg	500 mg	Dose per administration (mg)	Dose per administration (mg)
≤1.26	1500	-	3	1150	800
1.27 –1.38	1650	1	3	1300	800
1.39 –1.52	1800	2	3	1450	950
1.53 –1.66	2000	-	4	1500	1000
1.67 –1.78	2150	1	4	1650	1000
1.79 –1.92	2300	2	4	1800	1150
1.93 –2.06	2500	-	5	1950	1300
2.07 –2.18	2650	1	5	2000	1300
≥2.19	2800	2	5	2150	1450

Table 2: Standard and reduced dose calculations according to body surface area for a starting dose of CAPIIBINE of 1000 mg/m²

	Dose level 1000 mg/m²(twice daily)				
	Full dose 1000 mg/m ²	Number of tablets per administration (each administration to be given morning and evening)		Reduced dose (75%) 750 mg/m ²	Reduced dose (50%) 500 mg/m ²
Body surface area (m ²)	Dose per administration (mg)	150 mg	500 mg	Dose per administration (mg)	Dose per administration (mg)

≤1.26	1150	1	2	800	600
1.27 –1.38	1300	2	2	1000	600
1.39 –1.52	1450	3	2	1100	750
1.53 –1.66	1600	4	2	1200	800
1.67 –1.78	1750	5	2	1300	800
1.79 –1.92	1800	2	3	1400	900
1.93 –2.06	2000	-	4	1500	1000
2.07 –2.18	2150	1	4	1600	1050
≥ 2.19	2300	2	4	1750	1100

Dosage adjustments during treatment

General:

Toxicity due to CAPIIBINE administration may be managed by symptomatic treatment and/or modification of the CAPIIBINE dose (treatment interruption or dose reduction). Once the dose has been reduced it should not be increased at a later time.

For those toxicities considered by the treating physician to be unlikely to become serious or life-threatening treatment can be continued at the same dose without reduction or interruption.

Dosage modifications are not recommended for Grade 1 events. Therapy with CAPIIBINE should be interrupted if a Grade 2 or 3 adverse drug reaction (ADR) occurs. Once the ADR has resolved or decreased in intensity to Grade 1, CAPIIBINE therapy may be restarted at full dose or as adjusted according to Table 3. If a Grade 4 ADR occurs, therapy should be discontinued or interrupted until the ADR has resolved or decreased to Grade 1, and therapy should be restarted at 50% of the original dose. Patients taking CAPIIBINE should be informed of the need to interrupt treatment immediately if moderate or severe toxicity occurs. Doses of CAPIIBINE omitted for toxicity are not replaced.

Haematology:

Patients with baseline neutrophil counts of $<1.5 \times 10^9/L$ and/or thrombocyte counts of $<100 \times 10^9/L$ should not be treated with CAPIIBINE. If unscheduled laboratory assessments during a treatment cycle show grade 3 or 4 haematologic toxicity, treatment with CAPIIBINE should be interrupted.

The following table shows the recommended dose modifications following toxicity related to with CAPIIBINE:

Table 3: CAPIIBINE dose reduction schedule

Toxicity NCIC grades*	Dose changes within a treatment cycle	Dose adjustment for next cycle (% of starting dose)
Grade 1	maintain dose level	maintain dose level
Grade 2		
1 st appearance	interrupt until resolved to Grade 0-1	100%
2 nd appearance		75%
3 rd appearance		50%
4 th appearance	discontinue treatment permanently	Not applicable
Grade 3		
1 st appearance	interrupt until resolved to Grade 0-1	75%
2 nd appearance		50%
3 rd appearance	discontinue treatment permanently	Not applicable

Grade 4		
1 st appearance	discontinue permanently <i>or</i> If physician deems it to be in the patient's best interest to continue, interrupt until resolved to Grade	50%
2 nd appearance	discontinue permanently	Not applicable

* According to the National Cancer Institute of Canada Clinical Trial Group (NCIC CTG) Common Toxicity Criteria (version 1) or the Common Terminology Criteria for Adverse Events (CTCAE) of the Cancer Therapy Evaluation Program, US National Cancer Institute, version 3.0. For hand-foot syndrome and hyperbilirubinemia, (see section, Warnings and Precautions).

The following are the recommended dose modifications for toxicity when CAPIIBINE and docetaxel are used in combination:

Table 4: CAPIIBINE (X) in combination with docetaxel (T) dose reduction schedule

	Recommended Dose Modifications	
	CAPIIBINE dose changes within a treatment cycle	
Toxicity grade ¹	Grade 1	
	100% of starting dose (no interruption)	X: 100% of starting dose T: 100% (75 mg/m ²)
Toxicity grade ¹	Grade 2	
1 st appearance	Interrupt until resolved (grade 0 – 1)	X: 100% of starting dose T: 100% (75 mg/m ²)
2 nd appearance of same toxicity	Interrupt until resolved (grade 0 – 1)	X: 75% of starting dose T: Reduce to 55 mg/m ²
3 rd appearance of same toxicity	Interrupt until resolved (grade 0 – 1)	X: 50% of starting dose T: Discontinue permanently
4 th appearance of same toxicity	Discontinue permanently	
Toxicity grade	Grade 3	
	If grade 3 haematological see section on haematological toxicity, otherwise:	
1 st appearance	Interrupt until resolved (grade 0 – 1)	X: 75% of starting dose T: Reduce to 55 mg/m ²
2 nd appearance	Interrupt until resolved (grade 0 – 1)	X: 50% of starting dose T: Discontinue permanently
3 rd appearance	Discontinue permanently	
Toxicity grade ¹	Grade 4	
	If G4 haematological see section on haematological toxicity, otherwise:	
1 st appearance	Discontinue permanently <i>or</i> (if physician deems it to be in the best of interest of the patient) interrupt until resolved (grade 0-1)	X: Reduce to 50% T: Discontinue permanently
2 nd appearance	Discontinue permanently	

* National Cancer Institute of Canada Common Toxicity Criteria were used except for hand-foot syndrome (see Warnings and Precautions)

Specific dose adjustment in combination with docetaxel

CAPIIBINE and/or docetaxel dose modifications should be made according to the general dose modification scheme above, if nothing else is stated regarding specific dose adjustments. For those toxicities considered unlikely to become serious or life-threatening, e.g. alopecia, altered taste, nail changes, treatment can be continued at the same dose without reduction or interruption. At the beginning of a treatment cycle, if either a docetaxel or a CAPIIBINE treatment delay is indicated, both docetaxel and CAPIIBINE administration should be delayed until the requirements for restarting both drugs are met. If docetaxel has to be discontinued, CAPIIBINE treatment can be resumed when the requirements for restarting CAPIIBINE are met.

Hematology: Treatment should only be re-administered when the neutrophil count is $\geq 1.5 \times 10^9/l$ (Grade 0 - 1). Patients with neutropenia $< 0.5 \times 10^9/l$ (Grade 4) for more than 1 week, or febrile ($> 38^\circ C$) neutropenia, should have the docetaxel dosage reduced from 75 mg/m^2 to 55 mg/m^2 . If Grade 4 neutropenia or febrile neutropenia occurs at 55 mg/m^2 docetaxel, docetaxel should be discontinued. Patients with baseline neutrophil counts of $< 1.5 \times 10^9/l$ and/or thrombocyte counts of $< 1.0 \times 10^9/l$ should not be treated with the CAPIIBINE/docetaxel combination.

Hypersensitivity: Patients who develop severe hypersensitivity reactions (hypotension with a decrease of ≥ 20 mm Hg, or bronchospasm, or generalised rash/erythema) should stop treatment immediately and be given appropriate therapy. These patients should not be re challenged with the drug suspected to have caused hypersensitivity.

Peripheral neuropathy: For 1st appearance of Grade 2 toxicity, reduce the docetaxel dose to 55 mg/m^2 . If Grade 3 toxicity appears, discontinue docetaxel treatment. In both instances follow the above dose modification scheme for CAPIIBINE.

Fluid retention: Severe (Grade 3 or 4) toxicity such as pleural effusion, pericardial effusion or ascites which is possibly related to docetaxel should be closely monitored. In case of appearance of such toxicity docetaxel treatment should be discontinued, CAPIIBINE treatment may be continued without dose modification.

Hepatic impairment: Docetaxel should generally not be given to patients with serum bilirubin above the upper limit of normal. The following modifications should be applied to the docetaxel dose in the event of abnormal values for ASAT, ALAT, and/or alkaline phosphatase levels:

Table 5: Modifications to the docetaxel dose

ASAT and/or ALAT values	Alkaline phosphatase values	Docetaxel dose modification
$\leq 1.5 \times \text{UNL}$	and $\leq 5 \times \text{UNL}$	no dose modification
$> 1.5 \times \text{UNL} - \leq 2.5 \times \text{UNL}$	and $\leq 2.5 \times \text{UNL}$	no dose modification
$> 2.5 \times \text{UNL} - \leq 5 \times \text{UNL}$	and $\leq 5 \times \text{UNL}$	reduce by 25% (not below 55 mg/m^2)
$> 1.5 \times \text{UNL} - \leq 5 \times \text{UNL}$	and $> 2.5 \times \text{UNL} - \leq 5 \times \text{UNL}$	reduce by 25% (not below 55 mg/m^2)
$> 5 \times \text{UNL}$ (unless bone metastases are present in the absence of any liver disorder)	or $> 5 \times \text{UNL}$	delay dose by a maximum of 2 weeks. If no recovery, discontinue docetaxel.

Once the docetaxel dose is reduced for a given cycle, no further dose reduction is recommended for subsequent cycles unless worsening of the parameters is observed. In case of recovery of liver function tests after previous reduction of the docetaxel dose, the docetaxel dose can be re-escalated to the previous dose level.

Dehydration: Dehydration should be prevented or corrected at the onset. Patients with anorexia, asthenia, nausea, vomiting or diarrhea may rapidly become dehydrated. If Grade 2 (or higher) dehydration occur, CAPIIBINE treatment should be immediately interrupted and the dehydration corrected. Treatment should not be restarted until the patient is rehydrated and any precipitating causes have been corrected or controlled. Dose modifications applied should be those for the precipitating adverse event in accordance with the above guidelines.

Reductions to 75% and 50% of CAPIIBINE dose

For patients receiving CAPIIBINE monotherapy or CAPIIBINE in combination with docetaxel, the following tables show the dosage at 75% and 50%, calculated according to the body surface area:

Table 6: Calculated CAPIIBINE dose, reduced to 75% of the standard starting dose

Dose level 950 mg/m ² twice daily		Number of tablets administered in the morning		Number of tablets administered in the evening	
Body surface area (m ²)	Dose per administration (mg)	150 mg	500 mg	150 mg	500 mg
≤1.26	1150	1	2	1	2
1.27 – 1.38	1300	2	2	2	2
1.39 – 1.52	1450	3	2	3	2
1.53 – 1.66	1500	–	3	–	3
1.67 – 1.78	1650	1	3	1	3
1.79 – 1.92	1800	2	3	2	3
1.93 – 2.06	1950	3	3	3	3
2.07 – 2.18	2000	–	4	–	4
≥2.19	2150	1	4	1	4

Calculated CAPIIBINE dose, reduced to 50% of the standard starting dose

Dose level 625 mg/m ² twice daily		Number of tablets administered in the morning		Number of tablets administered in the evening	
Body surface area (m ²)	Dose per administration (mg)	150 mg	500 mg	150 mg	500 mg
≤1.38	800	2	1	2	1
1.39 – 1.52	950	3	1	3	1
1.53 – 1.66	1000	–	2	–	2
1.67 – 1.78	1000	–	2	–	2
1.79 – 1.92	1150	1	2	1	2
1.93 – 2.06	1300	2	2	2	2
2.07 – 2.18	1300	2	2	2	2
≥2.19	1450	3	2	3	2

Special dosage instructions

Pediatric use

The safety and efficacy of CAPIIBINE in children and adolescents (<18 years) have not been established.

Geriatric use

For CAPIIBINE monotherapy, no adjustment of the starting dose is needed. However, severe Grade 3 or 4 treatment-related ADRs were more frequent in patients over 80 years of age compared to younger patients. When CAPIIBINE was used in combination with other antineoplastic agents, geriatric patients (≥ 65 years) experience more Grade 3 and Grade 4 ADRs and ADRs that led to discontinuation, than younger patients. Careful monitoring of elderly patients is advisable.

In combination with docetaxel: an increased incidence of Grade 3 or 4 treatment-related ADRs and treatment-related serious ADRs was observed in patients 60 years of age or more. For patients 60 years of age or more treated with the combination of CAPIIBINE plus docetaxel, a starting dose reduction of CAPIIBINE to 75% (950 mg/m² twice daily) is recommended. For dosage calculations, see Table 1.

In combination with irinotecan: for patients 65 years of age or more, a starting dose reduction of CAPIIBINE to 800 mg/m² twice daily is recommended.

Renal impairment

In patients with moderate renal impairment (creatinine clearance 30 - 50 mL/min [Cockcroft and Gault]) at baseline, a dose reduction to 75% for a starting dose of 1250 mg/m² is recommended. In patients with mild renal impairment (creatinine clearance 51 - 80 mL/min), no adjustment in starting dose is recommended.

Careful monitoring and prompt treatment interruption is recommended if the patient develops a Grade 2, 3, or 4 ADRs with subsequent dose adjustment as outlined in Table 3 above (see also section Pharmacokinetics in Special Populations). If the calculated creatinine clearance decreases during treatment to a value below 30 mL/min, CAPIIBINE should be discontinued. The dose adjustment recommendations for patients with moderate renal impairment apply both to monotherapy and combination use. For dosage calculations, see Table 1 and Table 2.

Hepatic Impairment

In patients with mild to moderate hepatic impairment due to liver metastases, no starting dose adjustment is necessary. However, such patients should be carefully monitored (see section Pharmacokinetics in Special Populations and section, Warnings and Precautions). Patients with severe hepatic impairment have not been studied.

Mode of Administration

Oral

Contraindication

- Capecitabine is contraindicated in patients with known hypersensitivity to capecitabine or to any of its components.
- Capecitabine is contraindicated in patients who have history of severe and unexpected reactions to fluoropyrimidine therapy or with known hypersensitivity to fluorouracil.
- Capecitabine is contraindicated in patients with known complete absence of dihydropyrimidine dehydrogenase (DPD) activity
- Capecitabine should not be administered concomitantly with sorivudine or its chemically related analogues, such as brivudine.
- Capecitabine is contraindicated in patients with severe renal impairment (creatinine clearance below 30 mL/min).
- If contraindications exist to any of the agents in a combination regimen, that agent should not be used.

Warnings and precautions

Dose limiting toxicities. Dose limiting toxicities include diarrhoea, abdominal pain, nausea, stomatitis and hand-foot syndrome (hand-foot skin reaction, palmar-plantar erythrodysesthesia). Most adverse reactions are reversible and do not require permanent discontinuation of therapy, although doses may need to be withheld or reduced.

Diarrhoea. Patients with severe diarrhoea should be carefully monitored and given fluid and electrolyte replacement if they become dehydrated. Standard antidiarrhoeal treatments (e.g. loperamide) may be used. NCIC CTC grade 2 diarrhoea is defined as an increase of 4 to 6 stools/day or nocturnal stools, grade 3 diarrhoea as an increase of 7 to 9 stools/day or incontinence and malabsorption. Grade 4 diarrhoea is an increase of ≥ 10 stools/day or grossly bloody diarrhoea or the need for parenteral support. Dose reduction should be applied as necessary.

Dehydration. Dehydration should be prevented or corrected at the onset. Patients with anorexia, asthenia, nausea, vomiting or diarrhoea may rapidly become dehydrated. Dehydration may cause acute renal failure, especially in patients with pre-existing compromised renal function or when capecitabine is given concomitantly with known nephrotoxic medicinal products. Acute renal failure secondary to dehydration might be potentially fatal. If grade 2 (or higher) dehydration occurs, capecitabine treatment should be immediately interrupted and the dehydration corrected. Treatment should not be restarted until the patient is rehydrated and any precipitating causes have been corrected or controlled. Dose modifications applied should be applied for the precipitating adverse event as necessary.

Hand-foot syndrome (also known as hand-foot skin reaction or palmar-plantar erythrodysesthesia or chemotherapy induced acral erythema). Grade 1 hand- foot syndrome is defined as numbness, dysesthesia/paresthesia, tingling, painless swelling or erythema of the hands and/or feet and/or discomfort which does not disrupt the patient's normal activities.

Grade 2 hand- foot syndrome is painful erythema and swelling of the hands and/or feet and/or discomfort affecting the patient's activities of daily living.

Grade 3 hand- foot syndrome is moist desquamation, ulceration, blistering and severe pain of the hands and/or feet and/or severe discomfort that causes the patient to be unable to work or perform activities of daily living. Persistent or severe hand-foot syndrome (Grade 2 and above) can eventually lead to loss of fingerprints which could impact patient identification. If grade 2 or 3 hand- foot syndrome occurs, administration of capecitabine should be interrupted until the event resolves or decreases in intensity to grade 1. Following grade 3 hand-foot syndrome, subsequent doses of capecitabine should be decreased. When capecitabine and cisplatin are used in combination, the use of vitamin B6 (pyridoxine) is not advised for symptomatic or secondary prophylactic treatment of hand- foot syndrome, because of published reports that it may decrease the efficacy of cisplatin. There is some evidence that dexpantenol is effective for hand-foot syndrome prophylaxis in patients treated with Capecitabine.

Cardiotoxicity. Cardiotoxicity has been associated with fluoropyrimidine therapy, including myocardial infarction, angina, dysrhythmias, cardiogenic shock, sudden death and electrocardiographic changes (including very rare cases of QT prolongation). These adverse reactions may be more common in patients with a prior history of coronary artery disease. Cardiac arrhythmias (including ventricular fibrillation, torsade de pointes, and bradycardia), angina pectoris, myocardial infarction, heart failure and cardiomyopathy have been reported in patients receiving capecitabine. Caution must be exercised in patients with history of significant cardiac disease, arrhythmias and angina pectoris.

Hypo- or hypercalcaemia. Hypo- or hypercalcaemia has been reported during capecitabine treatment. Caution must be exercised in patients with pre-existing hypo- or hypercalcaemia.

Central or peripheral nervous system disease. Caution must be exercised in patients with central or peripheral nervous system disease, e.g. brain metastasis or neuropathy.

Diabetes mellitus or electrolyte disturbances. Caution must be exercised in patients with diabetes mellitus or electrolyte disturbances, as these may be aggravated during capecitabine treatment.

Coumarin-derivative anticoagulation. There was a significant increase in the mean AUC of S-warfarin. These results suggest an interaction, probably due to an inhibition of the cytochrome P450 2C9 isoenzyme system by capecitabine. Patients receiving concomitant capecitabine and oral coumarin-derivative anticoagulant therapy should have their anticoagulant response (INR or prothrombin time) monitored closely and the anticoagulant dose adjusted accordingly.

Brivudine. Brivudine must not be administered concomitantly with capecitabine. Fatal cases have been reported following this drug interaction. There must be at least a 4-week waiting period between end of treatment with brivudine and start of capecitabine therapy. Treatment with brivudine can be started 24 hours after the last dose of capecitabine. In the event of accidental administration of brivudine to patients being treated with capecitabine, effective measures should be taken to reduce the toxicity of capecitabine. Immediate admission to hospital is recommended. All measures should be initiated to prevent systemic infections and dehydration.

Hepatic impairment. Capecitabine use should be carefully monitored in patients with mild to moderate liver dysfunction, regardless of the presence or absence of liver metastasis. Administration of capecitabine should be interrupted if treatment-related elevations in bilirubin of $>3.0 \times \text{ULN}$ or treatment-related elevations in hepatic aminotransferases (ALT, AST) of $>2.5 \times \text{ULN}$ occur. Treatment with capecitabine monotherapy may be resumed when bilirubin decreases to $\leq 3.0 \times \text{ULN}$ or hepatic aminotransferases decrease to $\leq 2.5 \times \text{ULN}$.

Renal impairment. The incidence of grade 3 or 4 adverse reactions in patients with moderate renal impairment (creatinine clearance 30-50 ml/min) is increased compared to the overall population.

Dihydropyrimidine dehydrogenase (DPD) deficiency: DPD activity is rate limiting in the catabolism of 5-fluorouracil. Patients with DPD deficiency are therefore at increased risk of fluoropyrimidines-related toxicity, including for example stomatitis, diarrhoea, mucosal inflammation, neutropenia and neurotoxicity. DPD-deficiency related toxicity usually occurs during the first cycle of treatment or after dose increase.

Complete DPD deficiency. Complete DPD deficiency is rare (0.01-0.5% of Caucasians). Patients with complete DPD deficiency are at high risk of life-threatening or fatal toxicity and must not be treated with Capecitabine.

Partial DPD deficiency. Partial DPD deficiency is estimated to affect 3-9% of the Caucasian population. Patients with partial DPD deficiency are at increased risk of severe and potentially life-threatening toxicity. A reduced starting dose should be considered to limit this toxicity. DPD deficiency should be considered as a parameter to be taken into account in conjunction with other routine measures for dose reduction. Initial dose reduction may impact the efficacy of treatment. In the absence of serious toxicity, subsequent doses may be increased with careful monitoring.

Testing for DPD deficiency. Phenotype and/or genotype testing prior to the initiation of treatment with Capecitabine is recommended despite uncertainties regarding optimal pre-treatment testing methodologies. Consideration should be given to applicable clinical guidelines.

Genotypic characterisation of DPD deficiency. Pre-treatment testing for rare mutations of the DPYD gene can identify patients with DPD deficiency.

The four DPYD variants c.1905+1G>A [also known as DPYD*2A], c.1679T>G [DPYD*13], c.2846A>T

and c.1236G>A/HapB3 can cause complete absence or reduction of DPD enzymatic activity. Other rare variants may also be associated with an increased risk of severe or life-threatening toxicity.

Certain homozygous and compound heterozygous mutations in the DPYD gene locus (e.g. combinations of the four variants with at least one allele of c.1905+1G>A or c.1679T>G) are known to cause complete or near complete absence of DPD enzymatic activity.

Patients with certain heterozygous DPYD variants (including c.1905+1G>A, c.1679T>G, c.2846A>T and c.1236G>A/HapB3 variants) have increased risk of severe toxicity when treated with fluoropyrimidines.

The frequency of the heterozygous c.1905+1G>A genotype in the DPYD gene in Caucasian patients is around 1%, 1.1% for c.2846A>T, 2.6-6.3% for c.1236G>A/HapB3 variants and 0.07 to 0.1% for c.1679T>G.

Data on the frequency of the four DPYD variants in other populations than Caucasian is limited. At the present, the four DPYD variants (c.1905+1G>A, c.1679T>G, c.2846A>T and c.1236G>A/HapB3) are considered virtually absent in populations of African (-American) or Asian origin.

Phenotypic characterisation of DPD deficiency. For phenotypic characterisation of DPD deficiency, the measurement of pre-therapeutic blood levels of the endogenous DPD substrate uracil (U) in plasma is recommended.

Elevated pre-treatment uracil concentrations are associated with an increased risk of toxicity. Despite uncertainties on uracil thresholds defining complete and partial DPD deficiency, a blood uracil level ≥ 16 ng/ml and < 150 ng/ml should be considered indicative of partial DPD deficiency and associated with an increased risk for fluoropyrimidine toxicity. A blood uracil level ≥ 150 ng/ml should be considered indicative of complete DPD deficiency and associated with a risk for life-threatening or fatal fluoropyrimidine toxicity.

Ophthalmologic complications: Patients should be carefully monitored for ophthalmological complications such as keratitis and corneal disorders, especially if they have a prior history of eye disorders. Treatment of eye disorders should be initiated as clinically appropriate.

Severe skin reactions: Capecitabine can induce severe skin reactions such as Stevens-Johnson syndrome and Toxic Epidermal Necrolysis. Capecitabine should be permanently discontinued in patients who experience a severe skin reaction during treatment.

As this medicinal product contains anhydrous lactose as an excipient, patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Capecitabine tablets should not be crushed or cut. In case of exposure of either patient or caregiver to crushed or cut Capecitabine tablets adverse drug reactions could occur.

Interaction with other medicinal products and other forms of interactions

Interaction with other medicinal products

Brivudine: a clinically significant interaction between brivudine and fluoropyrimidines (e.g. capecitabine, 5-Fluorouracil, tegafur), resulting from the inhibition of dihydropyrimidine dehydrogenase by brivudine, has been described. This interaction, which leads to increased fluoropyrimidine toxicity, is potentially fatal. Therefore, brivudine must not be administered concomitantly with capecitabine. There must be at least a 4-week waiting period between end of treatment with brivudine and start of capecitabine therapy. Treatment with brivudine can be started 24 hours after the last dose of capecitabine.

Cytochrome P-450 2C9 substrates: Other than warfarin, there is no data of interaction between capecitabine and other CYP2C9 substrates. Care should be exercised when capecitabine is co-administered with 2C9 substrates (e.g. phenytoin). See also interaction with coumarin-derivative anticoagulants below

Coumarin-derivative anticoagulants: altered coagulation parameters and/or bleeding have been reported in patients taking capecitabine concomitantly with coumarin-derivative anticoagulants such as warfarin and phenprocoumon. These reactions occurred within several days and up to several months after initiating capecitabine therapy and, in a few cases, within one month after stopping capecitabine. After a single 20 mg dose of warfarin, capecitabine treatment has been shown to increase the AUC of S-warfarin with an increase in INR value. Since metabolism of R-warfarin was not affected, these results indicate that capecitabine down-regulates isozyme 2C9, but has no effect on isozymes 1A2 and 3A4. Patients taking coumarin-derivative anticoagulants concomitantly with capecitabine should be monitored regularly for alterations in their coagulation parameters (PT or INR) and the anti-coagulant dose adjusted accordingly.

Phenytoin: increased phenytoin plasma concentrations resulting in symptoms of phenytoin intoxication in single cases have been reported during concomitant use of capecitabine with phenytoin. Patients taking phenytoin concomitantly with capecitabine should be regularly monitored for increased phenytoin plasma concentrations.

Folinic acid/folic acid: Folinic acid has no major effect on the pharmacokinetics of capecitabine and its metabolites. However, folinic acid has an effect on the pharmacodynamics of capecitabine and its toxicity may be enhanced by folinic acid: the maximum tolerated dose (MTD) of capecitabine alone using the intermittent regimen is 3000 mg/m² per day whereas it is only 2000 mg/m² per day when capecitabine was combined with folinic acid (30 mg orally bid). The enhanced toxicity may be relevant when switching from 5-FU/LV to a capecitabine regimen. This may also be relevant with folic acid supplementation for folate deficiency due to the similarity between folinic acid and folic acid.

Antacid: the effect of an aluminium hydroxide and magnesium hydroxide-containing antacid on the pharmacokinetics of capecitabine was investigated. There was a small increase in plasma concentrations of capecitabine and one metabolite (5'-DFCR); there was no effect on the 3 major metabolites (5'-DFUR, 5-FU and FBAL).

Allopurinol: interactions with allopurinol have been observed for 5-FU; with possible decreased efficacy of 5-FU. Concomitant use of allopurinol with capecitabine should be avoided.

Interferon alpha: the MTD of capecitabine was 2000 mg/m² per day when combined with interferon alpha-2a (3 MIU/m² per day) compared to 3000 mg/m² per day when capecitabine was used alone.

Radiotherapy: the MTD of capecitabine alone using the intermittent regimen is 3000 mg/m² per day, whereas, when combined with radiotherapy for rectal cancer, the MTD of capecitabine is 2000 mg/m² per day using either a continuous schedule or given daily Monday through Friday during a 6-week course of radiotherapy.

Oxaliplatin: no clinically significant differences in exposure to capecitabine or its metabolites, free platinum or total platinum occurred when capecitabine was administered in combination with oxaliplatin or in combination with oxaliplatin and bevacizumab.

Bevacizumab: there was no clinically significant effect of bevacizumab on the pharmacokinetic parameters of capecitabine or its metabolites in the presence of oxaliplatin.

Food interaction:

Patients are always instructed to administer capecitabine within 30 minutes after a meal. Since current safety and efficacy data are based upon administration with food, it is recommended that capecitabine be administered with food. Administration with food decreases the rate of capecitabine absorption

Fertility, pregnancy and lactation

Women of childbearing potential/Contraception in males and females

Women of childbearing potential should be advised to avoid becoming pregnant while receiving treatment with capecitabine. If the patient becomes pregnant while receiving capecitabine, the potential hazard to the foetus must be explained. An effective method of contraception should be used during treatment and for 6 months after the last dose of capecitabine.

Male patients with female partners of reproductive potential should use effective contraception during treatment and for 3 months following the last dose of capecitabine.

Pregnancy

There are no data in pregnant women using capecitabine; however, it should be assumed that capecitabine may cause foetal harm if administered to pregnant women. In reproductive toxicity data in animals, capecitabine administration caused embryoletality and teratogenicity. These findings are expected effects of fluoropyrimidine derivatives. Capecitabine is contraindicated during pregnancy.

Breast-feeding

It is not known whether capecitabine is excreted in human breast milk. No studies have been conducted to assess the impact of capecitabine on milk production or its presence in human breast milk. In lactating mice, considerable amounts of capecitabine and its metabolites were found in milk. As the potential for harm to the nursing infant is unknown, breast-feeding should be discontinued while receiving treatment with capecitabine and for 2 weeks after the final dose.

Fertility

There is no data on capecitabine and impact on fertility. The capecitabine pivotal studies included females of childbearing potential and males only if they agreed to use an acceptable method of birth control to avoid pregnancy for the duration of the study and for a reasonable period thereafter.

Effect on ability to drive and use machines

Capecitabine has minor or moderate influence on the ability to drive and use machines. Capecitabine may cause dizziness, fatigue and nausea.

Undesirable effects

Summary of the safety profile

The safety profiles of capecitabine monotherapy for the metastatic breast cancer, metastatic colorectal cancer and adjuvant colon cancer populations are comparable.

The most commonly reported and/or clinically relevant treatment-related adverse drug reactions (ADRs) were gastrointestinal disorders (especially diarrhoea, nausea, vomiting, abdominal pain, stomatitis), hand-foot syndrome (palmar-plantar erythrodysesthesia), fatigue, asthenia, anorexia, cardiotoxicity, increased renal dysfunction on those with preexisting compromised renal function, and thrombosis/embolism.

Tabulated list of adverse reactions

ADRs considered to be possibly, probably, or remotely related to the administration of capecitabine are listed in table 7 for capecitabine given as monotherapy and in table 8 for capecitabine given in combination with different chemotherapy regimens in multiple indications. The following headings are used to rank the ADRs by frequency: very common, common, uncommon, rare, very rare. Within each frequency grouping, ADRs are presented in order of decreasing seriousness.

Capecitabine Monotherapy:

Table 7 Summary of related ADRs reported in patients treated with capecitabine monotherapy

Body System	Very Common All grades	Common All grades	Uncommon Severe and/or Life-threatening (grade 3-4) or considered medically relevant	Rare/Very Rare (Post-Marketing Experience)
<i>Infections and infestations</i>	-	Herpes viral infection, Nasopharyngitis, Lower respiratory tract infection	Sepsis, Urinary tract infection, Cellulitis, Tonsillitis, Pharyngitis, Oral candidiasis, Influenza, Gastroenteritis, Fungal infection, Infection, Tooth abscess	
<i>Neoplasm benign, malignant and unspecified</i>	-	-	Lipoma	
<i>Blood and lymphatic system disorders</i>	-	Neutropenia, Anaemia	Febrile neutropenia, Pancytopenia, Granulocytopenia, Thrombocytopenia, Leukopenia, Haemolytic anaemia, International Normalised Ratio (INR) increased/Prothrombin time prolonged	
<i>Immune system disorders</i>	-	-	Hypersensitivity	
<i>Metabolism and nutrition disorders</i>	Anorexia	Dehydration, Weight decreased	Diabetes, Hypokalaemia, Appetite disorder, Malnutrition, Hypertriglyceridaemia,	
<i>Psychiatric disorders</i>	-	Insomnia, Depression	Confusional state, Panic attack, Depressed mood, Libido decreased	
<i>Nervous system disorders</i>	-	Headache, Lethargy, Dizziness, Parasthesia, Dysgeusia	Aphasia, Memory impairment, Ataxia, Syncope, Balance disorder, Sensory disorder, Neuropathy peripheral	Toxic leukoencephalopathy (very rare)

<i>Eye disorders</i>	-	Lacrimation increased, Conjunctivitis, Eye irritation	Visual acuity reduced, Diplopia	Lacrimal duct stenosis (rare), Corneal disorders(rare), keratitis (rare), punctate keratitis (rare)
<i>Ear and labyrinth disorders</i>	-	-	Vertigo, Ear pain	
<i>Cardiac disorders</i>	-	-	Angina unstable, Angina pectoris, Myocardial ischaemia/infarction, Atrial fibrillation, Arrhythmia, Tachycardia, Sinus tachycardia, Palpitations	Ventricular fibrillation (rare), QT prolongation (rare), Torsade de pointes (rare), Bradycardia (rare), Vasospasm (rare)
<i>Vascular disorders</i>	-	Thrombophlebitis	Deep vein thrombosis, Hypertension, Petechiae, Hypotension, Hot flush, Peripheral coldness	
<i>Respiratory, thoracic and mediastinal disorders</i>	-	Dyspnoea, Epistaxis, Cough, Rhinorrhoea	Pulmonary embolism, Pneumothorax, Haemoptysis, Asthma, Dyspnoea exertional	
<i>Gastrointestinal disorders</i>	Diarrhoea, Vomiting, Nausea, Stomatitis, Abdominal pain	Gastrointestinal haemorrhage, Constipation, Upper abdominal pain, Dyspepsia, Flatulence, Dry mouth	Intestinal obstruction, Ascites, Enteritis, Gastritis, Dysphagia, Abdominal pain lower, Oesophagitis, Abdominal discomfort, Gastrooesophageal reflux disease, Colitis, Blood in stool	
<i>Hepatobiliary disorders</i>	-	Hyperbilirubinemia, Liver function test abnormalities	Jaundice	Hepatic failure (rare), Cholestatic hepatitis (rare)
<i>Skin and subcutaneous tissue disorders</i>	Palmar-plantar erythro-dysaesthesia syndrome**	Rash, Alopecia, Erythema, Dry skin, Pruritus, Skin hyper-pigmentation, Rash macular, Skin desquamation,	Blister, Skin ulcer, Rash, Urticaria, Photosensitivity reaction, Palmar erythema, Swelling face, Purpura,	Cutaneous lupus erythematosus (rare), Severe skin reactions such as Stevens-Johnson Syndrome and toxic Epidermal

		Dermatitis, Pigmentation disorder, Nail disorder	Radiation recall syndrome	Necrolysis (very rare)
<i>Muskuloskeletal and connective tissue disorders</i>	-	Pain in extremity, Back pain, Arthralgia	Joint swelling, Bone pain, Facial pain, Musculoskeletal stiffness, Muscular weakness	
<i>Renal and urinary disorders</i>	-	-	Hydronephrosis, Urinary incontinence, Haematuria, Nocturia, Blood creatinine increased	
<i>Reproductive system and breast disorders</i>	-	-	Vaginal haemorrhage	
<i>General disorders and administration site conditions</i>	Fatigue, Asthenia	Pyrexia, Oedema peripheral, Malaise, Chest pain	Oedema, Chills, Influenza like illness, Rigors, Body temperature increased	

** Based on the post-marketing experience, persistent or severe palmar-plantar erythrodysesthesia syndrome can eventually lead to loss of fingerprints

Capecitabine in combination therapy:

Table 8 Summary of related ADRs reported in patients treated with capecitabine in combination treatment **in addition to** those seen with capecitabine monotherapy or seen at **a higher frequency grouping** compared to capecitabine monotherapy

Body System	Very common All grades	Common All grades	Rare/Very Rare (Post-Marketing Experience)
<i>Infections and infestations</i>	-	Herpes zoster, Urinary tract infection, Oral candidiasis, Upper respiratory tract infection, Rhinitis, Influenza, *Infection, Oral herpes	
<i>Blood and lymphatic system disorders</i>	*Neutropenia, *Leucopenia, *Anaemia, *Neutropenic fever, Thrombocytopenia	Bone marrow depression, *Febrile Neutropenia	
<i>Immune system disorders</i>	-	Hypersensitivity	

<i>Metabolism and nutrition disorders</i>	Appetite decreased	Hypokalaemia, Hyponatraemia, Hypomagnesaemia, Hypocalcaemia, Hyperglycaemia	
<i>Psychiatric disorders</i>	-	Sleep disorder, Anxiety	
<i>Nervous system disorders</i>	Paraesthesia, Dysaesthesia, Peripheral neuropathy, Peripheral sensory neuropathy, Dysgeusia, Headache	Neurotoxicity, Tremor, Neuralgia, Hypersensitivity reaction, Hypoaesthesia	
<i>Eye disorders</i>	Lacrimation increased	Visual disorders, Dry eye, Eye pain, Visual impairment, Vision blurred	
<i>Ear and labyrinth disorders</i>	-	Tinnitus, Hypoacusis	
<i>Cardiac disorders</i>	-	Atrial fibrillation, Cardiac ischaemia/infarction	
<i>Vascular disorders</i>	Lower limb oedema, Hypertension, ⁺ Embolism and thrombosis	Flushing, Hypotension, Hypertensive crisis, Hot flush, Phlebitis	
<i>Respiratory, thoracic and mediastinal system disorders</i>	Sore throat, Dysaesthesia pharynx	Hiccups, Pharyngolaryngeal pain, Dysphonia	
<i>Gastrointestinal disorders</i>	Constipation, Dyspepsia	Upper gastrointestinal haemorrhage, Mouth ulceration, Gastritis, Abdominal distension, Gastroesophageal reflux disease, Oral pain, Dysphagia, Rectal haemorrhage, Abdominal pain lower, Oral dysaesthesia, Paraesthesia oral, Hypoaesthesia oral, Abdominal discomfort	

<i>Hepatobiliary disorders</i>	-	Hepatic function abnormal	
<i>Skin and subcutaneous tissue disorders</i>	Alopecia, Nail disorder	Hyperhidrosis, Rash erythematous, Urticaria, Night sweats	
<i>Musculoskeletal and connective tissue disorders</i>	Myalgia, Arthralgia, Pain in extremity	Pain in jaw , Muscle spasms, Trismus, Muscular weakness	
<i>Renal and urinary disorder</i>	-	Haematuria, Proteinuria, Creatinine renal clearance decreased, Dysuria	Acute renal failure secondary to dehydration (rare)
<i>General disorders and administration site conditions</i>	Pyrexia, Weakness, ⁺ Lethargy, Temperature intolerance	Mucosal inflammation, Pain in limb, Pain, Chills, Chest pain, Influenza-like illness, ⁺ Fever, Infusion related reaction, Injection site reaction, Infusion site pain, Injection site pain	
<i>Injury, poisoning and procedural complications</i>	-	Contusion	

⁺ For each term, the frequency count was based on ADRs of all grades. For terms marked with a “+”, the frequency count was based on grade 3-4 ADRs. ADRs are added according to the highest incidence seen in any of the major combination trials.

Description of selected adverse reactions

Hand-foot syndrome:

For the capecitabine dose of 1250 mg/m² twice daily on days 1 to 14 every 3 weeks, a frequency of 53% to 60% of all-grades HFS was observed in capecitabine monotherapy and a frequency of 63% was observed in the capecitabine/docetaxel for the treatment of metastatic breast cancer. For the capecitabine dose of 1000 mg/m² twice daily on days 1 to 14 every 3 weeks, a frequency of 22% to 30% of all-grade HFS was observed in capecitabine combination therapy.

The following covariates were statistically significantly associated with an increased risk of developing HFS: increasing capecitabine starting dose (gram), decreasing cumulative capecitabine dose (0.1*kg), increasing relative dose intensity in the first six weeks, increasing duration of study treatment (weeks), increasing age (by 10 year increments), female gender, and good ECOG performance status at baseline (0 versus ≥1).

Diarrhoea:

Capecitabine can induce the occurrence of diarrhoea, which has been observed in patients.

The following covariates were statistically significantly associated with an increased risk of developing diarrhoea: increasing capecitabine starting dose (gram), increasing duration of study treatment (weeks), increasing age (by 10 year increments), and female gender. The following covariates were statistically significantly associated with a decreased risk of developing diarrhoea: increasing cumulative capecitabine dose ($0.1 \times \text{kg}$) and increasing relative dose intensity in the first six weeks.

Cardiotoxicity):

In addition to the ADRs described in tables 7 and 8, the following ADRs were associated with the use of capecitabine monotherapy: cardiomyopathy, cardiac failure, sudden death, and ventricular extrasystoles.

Encephalopathy:

In addition to the ADRs described in tables 7 and 8, encephalopathy was also associated with the use of capecitabine monotherapy .

Exposure to crushed or cut capecitabine tablets:

In the instance of exposure to crushed or cut capecitabine tablets, the following adverse drug reactions have been reported: eye irritation, eye swelling, skin rash, headache, paresthesia, diarrhoea, nausea, gastric irritation, and vomiting.

Special populations

Elderly patients:

It has been shown an increase in the incidence of treatment-related grade 3 and 4 adverse reactions and treatment-related serious adverse reactions with patients ≥ 60 years of age compared to patients < 60 years of age. Patients ≥ 60 years of age treated with capecitabine plus docetaxel also had more early withdrawals from treatment due to adverse reactions compared to patients < 60 years of age. Further, with increasing age (by 10 year increments) it was statistically significantly associated with an increased risk of developing HFS and diarrhoea and with a decreased risk of developing neutropenia.

Gender

Female gender was statistically significantly associated with an increased risk of developing HFS and diarrhoea and with a decreased risk of developing neutropenia.

Patients with renal impairment:

It has been shown an increase in the incidence of treatment-related grade 3 and 4 adverse reactions compared to patients with normal renal function. Patients with moderately impaired renal function show an increased rate of dose reduction and patients with no or mild renal impairment and an increase in early withdrawals from treatment and in patients with no or mild renal impairment.

Overdose and treatment

The manifestations of acute overdose include nausea, vomiting, diarrhoea, mucositis, gastrointestinal irritation and bleeding, and bone marrow depression. Medical management of overdose should include customary therapeutic and supportive medical interventions aimed at correcting the presenting clinical manifestations and preventing their possible complications.

Pharmacological Properties

Pharmacodynamic properties

Capecitabine is a non-cytotoxic fluoropyrimidine carbamate, which functions as an orally administered

precursor of the cytotoxic moiety 5-fluorouracil (5-FU). Capecitabine is activated via several enzymatic steps. The enzyme involved in the final conversion to 5-FU, thymidine phosphorylase (ThyPase), is found in tumour tissues, but also in normal tissues, albeit usually at lower levels. In human cancer xenograft models capecitabine demonstrated a synergistic effect in combination with docetaxel, which may be related to the upregulation of thymidine phosphorylase by docetaxel.

There is evidence that the metabolism of 5-FU in the anabolic pathway blocks the methylation reaction of deoxyuridylic acid to thymidylic acid, thereby interfering with the synthesis of deoxyribonucleic acid (DNA). The incorporation of 5-FU also leads to inhibition of RNA and protein synthesis. Since DNA and RNA are essential for cell division and growth, the effect of 5-FU may be to create a thymidine deficiency that provokes unbalanced growth and death of a cell. The effects of DNA and RNA deprivation are most marked on those cells which proliferate more rapidly and which metabolise 5-FU at a more rapid rate.

Pharmacokinetic properties

The pharmacokinetics of capecitabine have been evaluated over a dose range of 502-3514 mg/m²/day. The parameters of capecitabine, 5'-deoxy-5-fluorocytidine (5'-DFCR) and 5'-deoxy-5-fluorouridine (5'-DFUR) measured on days 1 and 14 were similar. The AUC of 5-FU was 30%-35% higher on day 14. Capecitabine dose reduction decreases systemic exposure to 5-FU more than dose-proportionally, due to non-linear pharmacokinetics for the active metabolite.

Absorption

After oral administration, capecitabine is rapidly and extensively absorbed, followed by extensive conversion to the metabolites, 5'-DFCR and 5'-DFUR. Administration with food decreases the rate of capecitabine absorption, but only results in a minor effect on the AUC of 5'-DFUR, and on the AUC of the subsequent metabolite 5-FU. At the dose of 1250 mg/m² on day 14 with administration after food intake, the peak plasma concentrations (C_{max} in µg/ml) for capecitabine, 5'-DFCR, 5'-DFUR, 5-FU and FBAL were 4.67, 3.05, 12.1, 0.95 and 5.46 respectively. The time to peak plasma concentrations (T_{max} in hours) were 1.50, 2.00, 2.00, 2.00 and 3.34. The AUC_{0-∞} values in µg·h/ml were 7.75, 7.24, 24.6, 2.03 and 36.3.

Distribution

Capecitabine, 5'-DFCR, 5'-DFUR and 5-FU are 54%, 10%, 62% and 10% protein bound, mainly to albumin in *In vitro* human plasma.

Biotransformation

Capecitabine is first metabolised by hepatic carboxylesterase to 5'-DFCR, which is then converted to 5'-DFUR by cytidine deaminase, principally located in the liver and tumour tissues. Further catalytic activation of 5'-DFUR then occurs by thymidine phosphorylase (ThyPase). The enzymes involved in the catalytic activation are found in tumour tissues but also in normal tissues, albeit usually at lower levels. The sequential enzymatic biotransformation of capecitabine to 5-FU leads to higher concentrations within tumour tissues. In the case of colorectal tumours, 5-FU generation appears to be in large part localised in tumour stromal cells. Following oral administration of capecitabine to patients with colorectal cancer, the ratio of 5-FU concentration in colorectal tumours to adjacent tissues was 3.2 (ranged from 0.9 to 8.0). The ratio of 5-FU concentration in tumour to plasma was 21.4 (ranged from 3.9 to 59.9, n=8) whereas the ratio in healthy tissues to plasma was 8.9 (ranged from 3.0 to 25.8, n=8). Thymidine phosphorylase activity was measured and found to be 4 times greater in primary colorectal tumour than in adjacent normal tissue. According to immune-histochemical data, thymidine phosphorylase appears to be in large part localised in tumour stromal cells.

5-FU is further catabolised by the enzyme dihydropyrimidine dehydrogenase (DPD) to the much less toxic dihydro-5-fluorouracil (FUH2). Dihydropyrimidinase cleaves the pyrimidine ring to yield 5-

fluoro-ureidopropionic acid (FUPA). Finally, β -ureido-propionase cleaves FUPA to α -fluoro- β -alanine (FBAL) which is cleared in the urine. Dihydropyrimidine dehydrogenase (DPD) activity is the rate limiting step. Deficiency of DPD may lead to increased toxicity of capecitabine.

Elimination

The elimination half-life ($t_{1/2}$ in hours) of capecitabine, 5'-DFCR, 5'-DFUR, 5-FU and FBAL were 0.85, 1.11, 0.66, 0.76 and 3.23 respectively. Capecitabine and its metabolites are predominantly excreted in urine; 95.5% of administered capecitabine dose is recovered in urine. Faecal excretion is minimal (2.6%). The major metabolite excreted in urine is FBAL, which represents 57% of the administered dose. About 3% of the administered dose is excreted in urine unchanged.

Combination therapy

No effect by capecitabine on the pharmacokinetics of docetaxel or paclitaxel (C_{max} and AUC) and no effect by docetaxel or paclitaxel on the pharmacokinetics of 5'-DFUR.

Pharmacokinetics in special populations

A population pharmacokinetic analysis was carried out after capecitabine treatment of 505 patients with colorectal cancer dosed at 1250 mg/m^2 twice daily. Gender, presence or absence of liver metastasis at baseline, Karnofsky Performance Status, total bilirubin, serum albumin, ASAT and ALAT had no statistically significant effect on the pharmacokinetics of 5'-DFUR, 5-FU and FBAL.

Patients with hepatic impairment due to liver metastases. According to a pharmacokinetic study in cancer patients with mild to moderate liver impairment due to liver metastases, the bioavailability of capecitabine and exposure to 5-FU may increase compared to patients with no liver impairment. There are no pharmacokinetic data on patients with severe hepatic impairment.

Patients with renal impairment. Based on a pharmacokinetic study in cancer patients with mild to severe renal impairment, there is no evidence for an effect of creatinine clearance on the pharmacokinetics of intact drug and 5-FU. Creatinine clearance was found to influence the systemic exposure to 5'-DFUR (35% increase in AUC when creatinine clearance decreases by 50%) and to FBAL (114% increase in AUC when creatinine clearance decreases by 50%). FBAL is a metabolite without anti-proliferative activity.

Elderly. Based on the population pharmacokinetic analysis, which included patients with a wide range of ages (27 to 86 years) and included 234 (46%) patients greater or equal to 65, age has no influence on the pharmacokinetics of 5'-DFUR and 5-FU. The AUC of FBAL increased with age (20% increase in age results in a 15% increase in the AUC of FBAL). This increase is likely due to a change in renal function.

PHARMACEUTICAL INFORMATION

Shelf Life

24 Months

Storage conditions

Store below 30°C

Precautions for Storage

Keep out of reach of children.

Nature and contents of container

Dosage form: Tablets

Packaging: Alu-PVC/PVDC Blister Pack

Capecitabine Film-Coated tablets USP 150 mg: 60's tablets (6 x 10's Blister) in an outer carton with a pack insert

Capecitabine Film-Coated tablets USP 500 mg: 120's tablets (12 x 10's Blister) in an outer carton with a pack insert

Manufactured in India by:

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